Exercise 1

My completed code is located inside **rigidbody2d_friction.py.**

Exercise 2

2a)

The slight rocking/sliding is a numerical artifact of the contact solver. We treat each contact independently and then average, rather than solving all contacts simultaneously, so the normal/tangential impulses do not cancel perfectly. With gravity on, each frame injects a small downward velocity, and the discrete integration plus position correction introduces tiny residual errors. Those errors appear as jitter and slow drift even when the theoretical static-friction condition is satisfied.

2b)

For an incline of 13° (0.227 rad), the theoretical threshold for no acceleration is mu_s = tan(13°) ~ 0.231. When mu_s is well above this value, the box should be stationary. In the simulation, however, it still jitters/drifts at a constant speed due to numerical issues discussed in 2a. It was observed that drift is present for coefficients up to 0.99 (although, at this point, the drift is barely perceptible).

As mu_s approaches ~0.23 from above, the drift becomes more noticeable. Once mu_s drops below ~0.23, the box should transition to clear acceleration down the plane rather than a slow constant-speed slide. The simulation reproduces this expected behaviour.

2c)

My code for this exercise is written in **rigidbody2d_friction_extra.py.**

I improved stability by adding a simple sleep/resting state and retaining full tangential-position correction only when the contact is within the static-friction domain. I track small velocities over several frames, and once both linear and angular velocities remain below a small threshold for a few frames, I freeze the body by manually setting its velocity to zero and disabling gravity for that body. This removes residual jitter caused by gravity nudging the contact every frame. Additionally, when the contact is sliding, I apply a scaled tangential position correction so the block can still move; when it's static, I fully undo tangential drift. These implementations reduce rocking/sliding at the critical angle without breaking normal sliding behaviour.

Exercise 3

3a)

My code for this exercise is written in **rigidbody2d_rolling.py**

Exercise 4

The code in Exercise 3 is implemented so that pressing the “N” key restarts the race and changes the scenario. We will complete the race without friction. The results are shown below:

raceMode=0 (hollow cylinder)
finish speed |v| = 2.433307
raceMode=1 (solid cylinder)
finish speed |v| = 2.821543
raceMode=2 (sphere)
finish speed |v| = 2.911135
raceMode=3 (sliding particle)
finish speed |v| = 3.116057

As we can see, the ordering of the race, from slowest to fastest, is:

1. Sliding particle (no rotation) is fastest.
2. Sphere (I = 2/5 MR^2)
3. Solid cylinder (I = 1/2 MR^2)
4. Hollow cylinder (I = MR^2) is the slowest.

This is the ordering we expect based on the moments of inertia provided in the assignment handout.